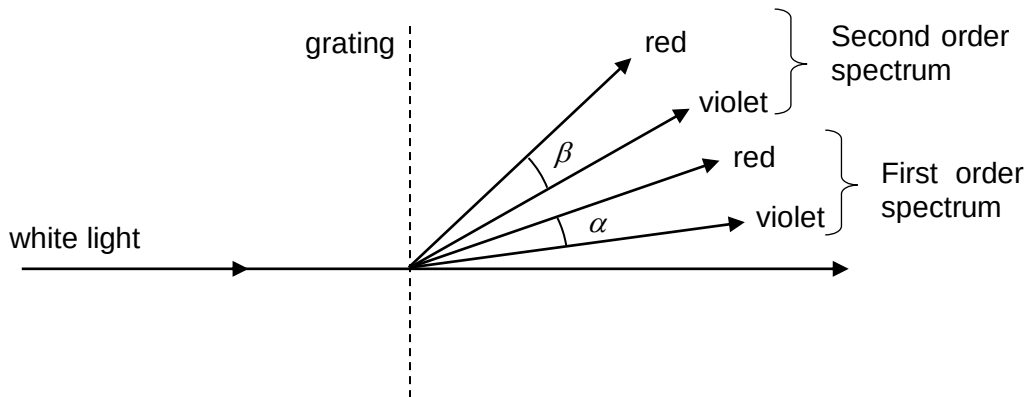


- 19 A diffraction grating with 6.0×10^5 lines per metre is placed at right angles to a ray of white light. The first and second order spectra produced are shown in the diagram.



The angle between the red and violet ends of the spectrum is α for the first order spectrum and β for the second order spectrum. How do α and β compare?

- A** $\alpha < \frac{1}{2} \beta$ **B** $\alpha = \frac{1}{2} \beta$ **C** $\alpha = \beta$ **D** $\alpha > \beta$

Ans: A

Wavelengths of white light range from 700 nm (red) to 400 nm (violet).

$$d \sin \theta = n\lambda$$

$$\alpha = \sin^{-1} (\lambda_{\text{red}} \times 6.0 \times 10^5) - \sin^{-1} (\lambda_{\text{violet}} \times 6.0 \times 10^5) = 11^\circ$$

$$\beta = \sin^{-1} (2\lambda_{\text{red}} \times 6.0 \times 10^5) - \sin^{-1} (2\lambda_{\text{violet}} \times 6.0 \times 10^5) = 28^\circ$$

- 20 In which of the following cases does an electric field do positive work on a charged particle?